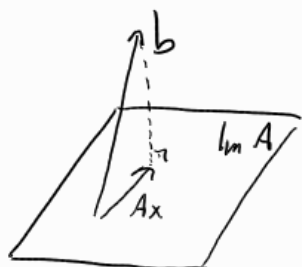


Linearni problem najm. kvadratov

$$Ax=b; A \in \mathbb{R}^{m \times n}; \text{rank}(A)=n; m \gg n$$

- v splošnem ne moremo zadostiti vsem enačbam



• najboljši fit $Ax \rightarrow b \perp \text{Im } A$



$$A^T A x = A^T b$$

← sistema ne želimo reševati, ker je občutljivost $A^T A$ večja od A
($\kappa(A^T A) = \kappa(A)^2$)

→ QR razcep:

$$\left. \begin{array}{l} 1) \cdot Q \in \mathbb{R}^{m \times n}; \text{ z ortonormiranimi stolpci} \\ \cdot R \in \mathbb{R}^{n \times n}; \text{ zg. trikotna} \end{array} \right\} Rx = Q^T b$$

$$A = Q \begin{bmatrix} R \\ 0 \end{bmatrix}$$

Razširjen QR razcep

$$\left. \begin{array}{l} 2) \cdot Q \in \mathbb{R}^{m \times m}; \text{ ortogonalna} \\ \cdot R \in \mathbb{R}^{m \times n}; \text{ kvazi-zg. trikotna} \end{array} \right\} R_1 x = C_1$$

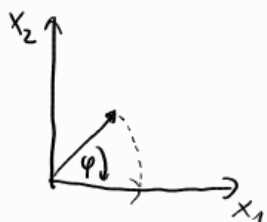
$$A = Q \begin{bmatrix} R \\ 0 \end{bmatrix}; R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$$

Razcepi:

1) Gram-Schmidtova ortogonalizacija

$$Q^T b = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

2) Givensove rotacije



$$R_{12}^T = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix}; \varphi = \arctan \frac{x_2}{x_1} \xrightarrow{\text{hi uporabno}} \text{raje prek skal. pr.}$$

$\xrightarrow{\text{v } \mathbb{R}^n}$ R_{ik} je rotacija v ravnini (i, k) ter identiteta drugje: $R_{ik} = \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & c & s \\ & & -s & c \\ & & & \ddots & \\ & & & & 1 \end{bmatrix}$

R_{ik} se imenuje Givensova rotacija

$$x \rightarrow y = R_{ik}^T x; \text{ želimo } y_k = 0$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_i \\ \vdots \\ x_k \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} y_1 \\ \vdots \\ y_k = 0 \\ \vdots \\ y_n \end{bmatrix}$$

$$r = \sqrt{x_i^2 + x_k^2} \rightarrow \begin{aligned} \cos \varphi &= x_i / r \\ \sin \varphi &= x_k / r \end{aligned}$$

→ QR razcep dobimo z:

$$\underbrace{R_{n,n}^T \dots R_{2,n}^T \dots R_{23}^T R_{1n}^T \dots R_{12}^T}_{Q^T} A = R \Rightarrow A = QR$$

3) Householderjeva zrcaljenja

Singularni razcep

$A \in \mathbb{R}^{m \times n}$, $m \geq n$; $A = U \Sigma V^T$; $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ ortogonalni, $\Sigma \in \mathbb{R}^{m \times n}$ kvazidiagonalna
z elementi $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$. σ_j so singularne vrednosti, stolpci U so levi sing. vekt.,
stolpci V so desni sing. vekt.